

Understanding Alien Language : Inspired by Project Hail Mary

A Physics-Based Approach Through Fourier Analysis

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Chapter 1

Can Physics Become a Universal Language?

1.1 The First Contact Problem

One of the most fascinating questions in modern science concerns the possibility of communicating with an extraterrestrial civilization. The vastness of the observable universe, containing hundreds of billions of galaxies and an even larger number of planetary systems, makes it difficult to dismiss the possibility that intelligent life may have emerged elsewhere. If such a civilization exists and possesses technology capable of transmitting information across interstellar distances, an intriguing scientific challenge immediately arises: how can a message originating from an entirely unknown civilization be understood?

At first glance, this appears to be a problem of language. Human communication relies on shared conventions developed through history, culture, and experience. Every spoken language encodes meaning according to rules that are familiar only to those who have learned them. An extraterrestrial civilization would almost certainly possess neither our vocabulary nor our grammatical structure. Consequently, a direct linguistic interpretation of an alien message would be impossible.

However, before asking what the message means, a physicist asks a more fundamental question.

What is the physical nature of the received signal?

This seemingly simple question changes the entire perspective of the problem. Regardless of its origin or intended meaning, every transmitted message must ultimately be carried by a physical phenomenon. It may travel as electromagnetic radiation, pressure waves, fluctuations of plasma, or some other propagating disturbance governed by the laws of nature. Long before meaning can be extracted, the underlying physical structure of the signal must first be understood.

The purpose of this book is to investigate that first stage of understanding.

1.2 Why Physics Comes Before Language

Imagine that tomorrow a radio telescope receives a signal unlike anything previously observed. Suppose further that careful verification confirms the signal to be artificial rather than naturally occurring.

What should scientists do first?

Should they immediately attempt to translate the signal?

Should they search for words or grammar?

The answer is no.

Translation presupposes that the underlying physical structure of the signal has already been understood. Before discussing syntax, semantics, or information content, one must determine how the signal itself is constructed. This is a fundamentally physical question.

Physics concerns itself not with meaning, but with structure. It seeks regularity hidden beneath apparent complexity. Whether one studies the vibration of a violin string, the oscillation of an electromagnetic field, or the quantum state of an electron, the objective remains the same: identify the fundamental components from which the observed phenomenon is built.

This philosophy forms the central theme of the present work. Rather than attempting to interpret an unknown language, we first ask how an unknown signal can be analysed using the universal principles of mathematical physics.

1.3 Nature Speaks Through Oscillations

Oscillatory behaviour appears throughout physics.

A stretched string vibrates.

Air molecules oscillate to produce sound.

Electric and magnetic fields propagate as electromagnetic waves.

Quantum particles are described by wavefunctions whose evolution is governed by oscillatory phase factors.

Even thermal motion can often be understood through the collective vibrations of atoms.

Although these phenomena arise from very different physical systems, they possess a remarkable mathematical similarity. Each may be viewed as a superposition of elementary oscillations possessing different frequencies.

This observation suggests an important idea.

If an unknown civilization communicates by transmitting a physical signal, the signal itself must also obey the universal laws governing waves and oscillations. While the meaning encoded within the message may remain unknown, its physical construction can still be investigated.

The search for structure therefore begins not with language, but with oscillation.

1.4 The Importance of Representation

One of the deepest lessons of theoretical physics is that the same physical system may admit multiple mathematical descriptions.

The trajectory of a particle may be represented in position space or momentum space.

An electromagnetic field may be described either in the time domain or in the frequency domain.

A quantum state may be expanded in terms of position eigenstates, momentum eigenstates, or energy eigenstates depending upon the physical question being investigated.

None of these representations is more fundamental than the others. Instead, each emphasizes different aspects of the same underlying reality.

This idea is central to the present work.

Rather than studying an unknown sound signal solely as a function of time, we seek a representation that naturally reveals the oscillatory components hidden within it. The mathematical framework that accomplishes this transformation is known as Fourier analysis.

As will become apparent in the following chapters, Fourier analysis is not merely a computational technique. It is a change of perspective.

1.5 Choice of Physical Signal

Up to this point, we have deliberately avoided specifying the physical nature of the received signal. An advanced extraterrestrial civilization could, in principle, communicate through a variety of physical carriers. The transmitted information might be encoded in electromagnetic radiation, acoustic waves, optical pulses, magnetic field variations, or any other measurable phenomenon capable of propagating through an appropriate medium.

From the perspective of mathematical physics, the particular choice of carrier is not fundamental. What ultimately reaches the observer is the output of a detector—a measurable quantity that varies with time. Whether the detector is a radio antenna, an optical photodetector, a microphone, or another sensing device, the resulting measurement may always be represented as a time-dependent function,

$$x(t).$$

Consequently, the mathematical framework developed throughout this book is not restricted to sound alone. The principles of Fourier analysis apply equally to any physical signal whose amplitude evolves with time.

Nevertheless, every scientific investigation benefits from a concrete physical model. Throughout the remainder of this book, we shall adopt an acoustic signal as our representative example. This choice is motivated by two considerations.

First, sound provides one of the most intuitive examples of an oscillatory physical phenomenon, allowing the underlying mathematics to be visualized naturally.

Second, the inspiration for the present work originates from Andy Weir’s novel *Project Hail Mary*, in which meaningful communication between the characters Grace and Rocky is established through sound. While the communication described in the novel is fictional, it raises a fascinating scientific question:

If an unknown acoustic signal were received, how would a physicist begin to analyse it?

The purpose of this book is to answer that question using the language of mathematical physics.

Looking Ahead

The discussion so far has deliberately avoided introducing the mathematics of Fourier analysis.

Before decomposing an unknown signal into its constituent frequencies, we must first understand the physical quantity being analysed.

Accordingly, the next chapter begins with the fundamental physics of sound itself. We shall examine how sound is represented mathematically as a continuous function of time and why this representation, although natural, conceals much of the information contained within the signal.

Chapter 2

The Physics of Sound

Although the discussion that follows is presented using acoustic waves, the reader should remember that every derivation developed in this chapter is fundamentally a statement about time-dependent physical signals. An intelligent civilization may choose to communicate through electromagnetic radiation, optical pulses, acoustic waves, or any other measurable physical phenomenon capable of carrying information. Regardless of the physical carrier, the detector ultimately records a quantity that varies with time, and it is this mathematical object that becomes the subject of Fourier analysis.

Throughout the present work, we deliberately adopt acoustic waves as our representative physical system. This choice is motivated not only by the intuitive nature of sound as an oscillatory phenomenon, but also by the communication between Grace and Rocky in Andy Weir's *Project Hail Mary*, which serves as the original inspiration for this exploration. Sound is therefore chosen not because it is unique, but because it provides one of the clearest physical windows into the mathematics of Fourier analysis.

2.1 Sound as a Physical Wave

Before a signal can be analysed mathematically, it must first be understood as a physical phenomenon.

Sound is a mechanical wave that propagates through a material medium by means of successive compressions and rarefactions of its constituent particles. Unlike electromagnetic waves, sound cannot propagate through vacuum because its transmission requires interactions between neighbouring particles of the medium.

At every point in space, the pressure of the medium fluctuates slightly about its equilibrium value. If the ambient atmospheric pressure is denoted by P_0 , then the instantaneous pressure may be written as

$$P(\mathbf{r}, t) = P_0 + \Delta P(\mathbf{r}, t),$$

where $\Delta P(\mathbf{r}, t)$ represents the pressure fluctuation produced by the propagating acoustic wave.

It is these pressure fluctuations, rather than the constant atmospheric pressure itself, that

carry the acoustic information from the source to the observer.

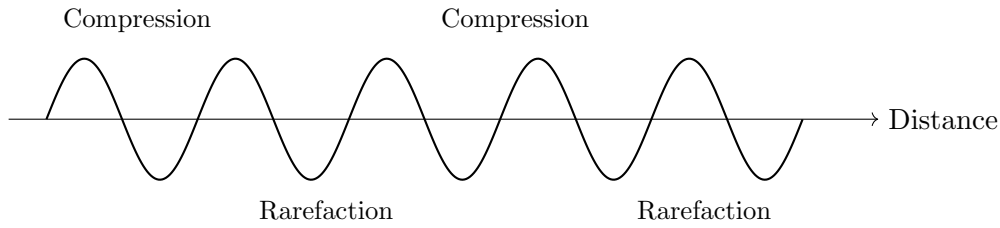


Figure 2.1: Pressure variation produced by a propagating acoustic wave. Regions of compression correspond to positive pressure fluctuations, while rarefaction corresponds to negative pressure fluctuations.

2.2 The Microphone as a Physical Detector

Suppose that the propagating acoustic wave reaches a microphone positioned at a fixed location.

Since the microphone remains stationary throughout the measurement, the spatial coordinates become fixed and only the temporal variation of the pressure remains relevant. Consequently,

$$P(\mathbf{r}, t) \longrightarrow P(t).$$

The microphone does not recognize words, language, or meaning. It performs only one physical task: it converts pressure fluctuations into a continuously varying electrical voltage. The diaphragm oscillates in response to the incident sound wave, and this mechanical motion is transformed into an electrical signal whose amplitude follows the incoming pressure variation.

Thus, before any attempt can be made to interpret the message, the observer possesses only a continuously varying physical quantity.

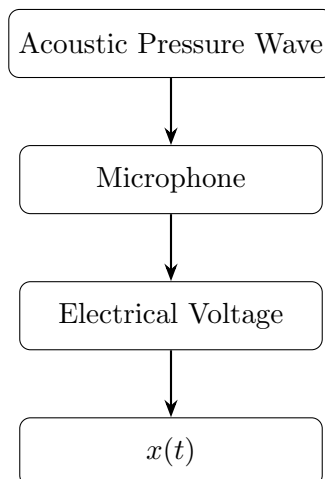


Figure 2.2: Transformation from a physical acoustic wave into the measurable mathematical signal $x(t)$.

2.3 The Mathematical Description of the Acoustic Signal

To describe the measured signal mathematically, we introduce the function

$$x(t),$$

where the independent variable t denotes time and the function value represents the instantaneous pressure variation recorded by the microphone.

More precisely,

$$x(t) = \Delta P(t),$$

where

$$\Delta P(t) = P(t) - P_0,$$

with P_0 denoting the ambient atmospheric pressure.

Throughout the remainder of this work, the symbol $x(t)$ will represent the received acoustic signal.

Every vibration of the microphone diaphragm and every piece of physical information carried by the incoming sound wave is encoded within this single function.

2.4 What Does the Signal Look Like?

The function $x(t)$ may be visualized as a continuously varying curve whose amplitude changes with time.

Every point on this curve corresponds to the pressure fluctuation measured by the microphone at a particular instant. Peaks represent regions of compression, whereas troughs correspond to regions of rarefaction in the propagating sound wave.

Although the waveform appears smooth and continuous, it represents nothing more than the physical evolution of the acoustic pressure with time.

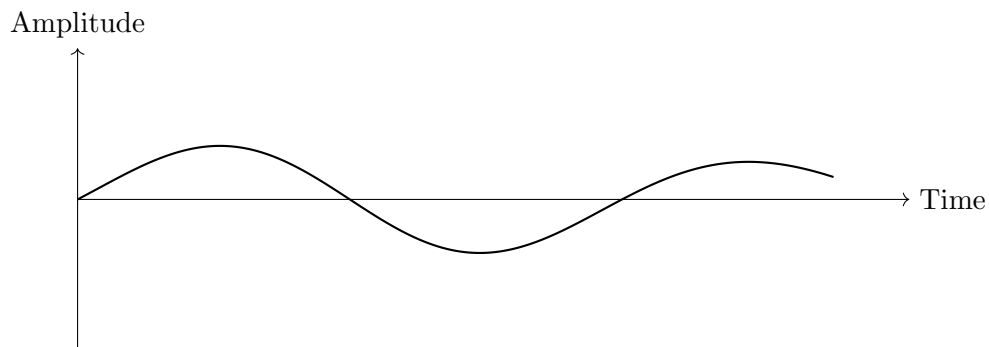


Figure 2.3: Illustrative acoustic waveform represented as a continuous function of time. The waveform shown is schematic and represents a generic time-domain signal rather than a pure sinusoid.

2.5 From Physical Measurement to Numerical Data

The waveform shown in the previous section provides an intuitive visual description of the acoustic signal. However, experimental measurements may also be represented numerically.

Each measurement corresponds to the pressure fluctuation recorded at a particular instant of time. Consequently, the same physical signal may be expressed as a sequence of measured values arranged in tabular form.

The graphical representation emphasizes the continuous evolution of the sound wave, whereas the numerical representation emphasizes the measured data from which the graph is constructed.

Although these descriptions appear different, they represent exactly the same physical phenomenon.

Table 2.1: Illustrative measurements of an acoustic signal.

Time (s)	Pressure Variation $x(t)$
0.000000	0.0012
0.000023	0.0018
0.000045	0.0025
0.000068	0.0016
0.000091	-0.0007

2.6 The Time-Domain Representation

Representing a signal as a function of time is known as the time-domain representation.

The time-domain description records every pressure fluctuation experienced by the microphone and therefore contains the complete physical information carried by the acoustic wave.

Nothing has been discarded.

Every oscillation, every variation in amplitude, and every subtle feature of the sound remains present within the function $x(t)$.

The challenge therefore is not the absence of information.

The challenge is that the underlying structure of the signal is not immediately visible.

Nature presents us with a continuously evolving physical phenomenon.

The graph is only our visual interpretation.

The table is only our numerical representation.

Neither is the signal itself.

The signal is the underlying physical process that both descriptions attempt to describe.

2.7 The Limitation of the Time-Domain Picture

Suppose two musical notes are played simultaneously.

The microphone faithfully records the resulting pressure fluctuations, producing the function $x(t)$. However, a visual inspection of the waveform alone rarely reveals that two independent oscillations are present.

The same difficulty would arise if an unknown civilization transmitted a complex acoustic message. Although the recorded waveform contains all of the physical information, its internal oscillatory structure remains hidden within the complicated time-domain signal.

The problem therefore is not that information is absent.

The problem is that the information is hidden.

Looking Ahead

This observation naturally raises an important question.

If complicated sound signals are built from simpler oscillations, what exactly is a simple oscillation?

Can every acoustic signal be understood as a combination of such elementary building blocks?

To answer these questions, we must first understand the fundamental mathematical description of oscillatory motion. This journey begins with the simplest oscillation found throughout physics: the simple harmonic wave.

"Every wave we observe in nature—whether a sound wave in air, a ripple on water, a vibrating string, or an electromagnetic wave in space—is ultimately an expression of oscillatory motion. The details of the medium may change, but the underlying principle remains remarkably universal: disturbances propagate because local oscillators interact with one another. If oscillations are the fundamental language in which nature expresses waves, then perhaps understanding any wave begins by identifying the oscillations from which it is composed. This idea is the conceptual foundation of Fourier analysis".

Chapter 3

Oscillations: The Natural Basis of Waves

The previous chapter established that every acoustic signal reaching the observer may be represented by a continuous function of time,

$$x(t).$$

Although this function contains the complete physical information carried by the sound wave, its internal structure is not immediately apparent. The waveform often appears highly complicated, concealing the simpler oscillatory processes responsible for its formation.

This observation naturally raises a fundamental question.

Can a complicated signal be understood by expressing it in terms of simpler, more elementary oscillations?

The answer to this question lies at the heart of wave physics and ultimately leads to one of the most powerful ideas in mathematical physics: the representation of an unknown object in an appropriate basis (remember this line for Future Q.M analogy).

3.1 The Simplest Oscillation in Nature

Many physical systems, when displaced slightly from equilibrium, exhibit oscillatory motion.

A vibrating guitar string, the oscillation of air molecules inside a musical instrument, the motion of a pendulum for small angular displacements, and the vibration of atoms within a crystal all possess one common characteristic.

They repeatedly return toward an equilibrium configuration.

Among all oscillatory motions, one occupies a particularly important position in physics.

This is the simple harmonic oscillation.

Its importance does not arise merely from its mathematical simplicity, but from the remarkable fact that more complicated oscillations may often be understood as combinations of these elementary motions.

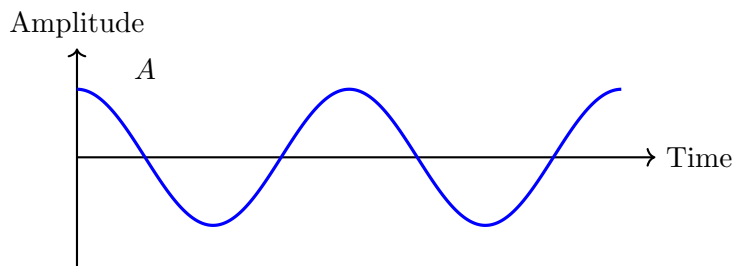


Figure 3.1: A simple harmonic oscillation described by $x(t) = A \cos(\omega t + \phi)$.

The mathematical description of a simple harmonic oscillation is

$$x(t) = A \cos(\omega t + \phi).$$

Each parameter possesses a clear physical interpretation.

The constant A determines the maximum amplitude of the oscillation.

The angular frequency ω specifies how rapidly the oscillation repeats.

The quantity ϕ represents the initial phase and determines the position of the oscillation at the instant $t = 0$.

Together, these three quantities completely characterize a single pure oscillation.

Nature rarely produces perfectly pure oscillations.

Nevertheless, pure oscillations occupy a special position in physics because they serve as the elementary building blocks from which more complicated waveforms can be constructed.

Much as individual vectors span a vector space, pure oscillations span the space of oscillatory signals.

3.2 Frequency: The Fingerprint of an Oscillation

Although every simple harmonic oscillation possesses the same mathematical form, not all oscillations are identical.

Some repeat slowly.

Others repeat extremely rapidly.

The physical quantity that distinguishes one oscillation from another is its frequency.

Frequency specifies the number of complete oscillations performed in one second and is measured in hertz (Hz), where

$$1 \text{ Hz} = 1 \text{ cycle per second.}$$

An oscillation of frequency 100 Hz completes one hundred cycles every second, whereas an oscillation of frequency 1000 Hz completes one thousand cycles during the same interval.

Thus, frequency serves as a unique identifier of an oscillatory mode.

Two oscillations possessing different frequencies represent physically distinct modes of vibration.

3.2.1 Period of Oscillation

An equivalent way of describing an oscillation is through its period, denoted by T . The period represents the time required for one complete oscillation.

Since one oscillation is completed every T seconds, the frequency and period are related by

$$f = \frac{1}{T}.$$

Thus, an oscillation with a shorter period possesses a higher frequency, while an oscillation with a longer period oscillates more slowly.

3.2.2 Angular Frequency

In physics, oscillatory motion is more naturally described in terms of angular displacement rather than the number of cycles. For this reason, we introduce the angular frequency,

$$\omega = 2\pi f,$$

which is measured in radians per second.

The equation of simple harmonic motion therefore assumes the familiar form

$$x(t) = A \cos(\omega t + \phi),$$

where the angular frequency determines how rapidly the oscillation evolves with time.

Physicists often find it more convenient to measure oscillations in terms of the angular displacement accumulated during one second rather than the number of cycles.

3.3 A Sound Wave Can Contain Many Frequencies

So far, we have considered only a single pure oscillation. Such a wave possesses one well-defined frequency and is described completely by a single cosine function.

Real acoustic signals, however, are rarely so simple.

Consider two pure tones, one having a frequency of 440 Hz and the other 880 Hz. Individually, they may be written as

$$x_1(t) = \cos(2\pi 440 t),$$

and

$$x_2(t) = \frac{1}{2} \cos(2\pi 880 t).$$

If both sounds are produced simultaneously, the microphone does not record two separate waveforms. Instead, it records only their combined pressure variation,

$$x(t) = \cos(2\pi 440 t) + \frac{1}{2} \cos(2\pi 880 t).$$

The resulting waveform is no longer a simple cosine. Instead, it represents the superposition of two distinct oscillatory modes having different frequencies.

This observation leads to a profound conclusion.

A complicated acoustic signal does not arise because Nature invents a new kind of oscillation. Rather, it arises because many simple oscillations coexist simultaneously, each contributing its own frequency to the overall waveform.

3.4 The Central Question

Suppose an unknown civilization transmits an acoustic message.

The microphone faithfully records the resulting waveform,

$$x(t).$$

The waveform undoubtedly contains all of the physical information carried by the signal. However, the individual frequencies responsible for producing that waveform remain hidden.

This immediately raises a fundamental question.

Given only the measured function $x(t)$, can we determine which frequencies are present and how strongly each one contributes to the signal?

Answering this question is precisely the objective of Fourier analysis.

In the next chapter, we shall develop the mathematical framework that makes such a decomposition possible.

A complicated waveform is therefore distinguished not by possessing a new kind of oscillation, but by containing many frequencies simultaneously.

If we could somehow determine the frequencies present within an unknown signal and measure the contribution of each one, we would gain a far deeper understanding of its physical structure than is possible by examining the waveform alone.

3.5 Complex Exponentials and Oscillatory Motion

At first sight, the introduction of complex numbers into the description of a real sound wave may appear unnecessary.

However, the complex exponential provides one of the most elegant representations of oscillatory motion in all of physics.

Euler's celebrated relation,

$$e^{i\theta} = \cos \theta + i \sin \theta,$$

demonstrates that every sinusoidal oscillation may be viewed as the real or imaginary part of a rotating complex vector.

Consequently,

$$e^{i\omega t}$$

may be regarded as the mathematical representation of a pure oscillatory mode possessing angular frequency ω .

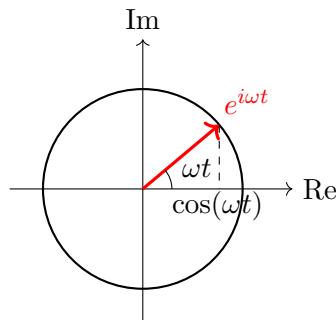


Figure 3.2: Geometric interpretation of Euler's formula. A rotating complex vector generates sinusoidal oscillations through its projection onto the real axis.

Suppose we possess an unknown acoustic signal.

Instead of attempting to interpret the entire waveform simultaneously, one may compare it with a single pure oscillation.

If the comparison is poor, that oscillation contributes very little.

If the comparison is strong, the corresponding oscillation plays an important role in constructing the signal.

Repeating this procedure for every possible frequency gradually reveals the internal oscillatory composition of the unknown waveform.

We have identified the elementary building blocks of oscillatory phenomena.

The remaining question is therefore obvious.

How can the contribution of each oscillatory basis function be measured quantitatively?

The mathematical answer to this question is provided by the Fourier Transform, which will be developed in the next chapter.

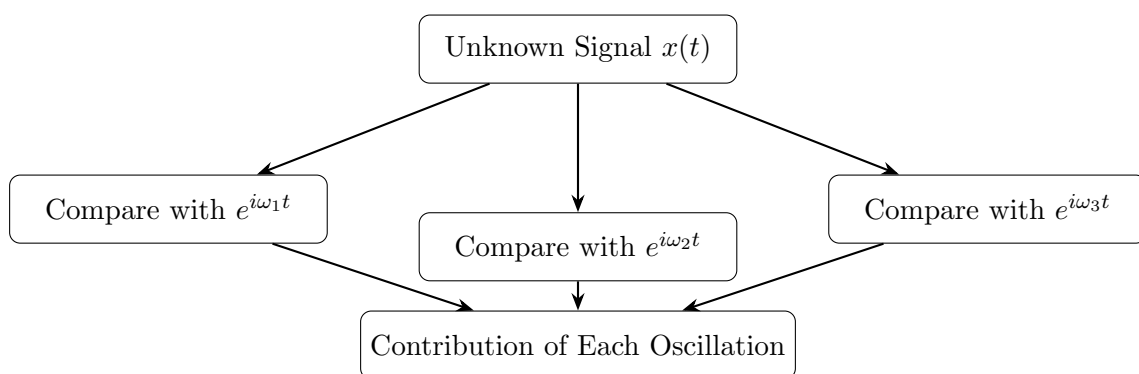


Figure 3.3: Conceptual illustration of analysing an unknown signal by comparing it with elementary oscillatory basis functions. This idea motivates the Fourier transform developed in the next chapter.

Chapter 4

Orthogonality: The Geometry Behind Fourier Analysis

The previous chapter established that every acoustic signal may be regarded as a superposition of many elementary oscillations, each distinguished by its own characteristic frequency.

This conclusion immediately raises a more fundamental question.

How do we mathematically compare two oscillations?

Suppose an unknown acoustic signal contains a frequency of 440 Hz. How can we determine that this particular oscillation is actually present within the measured waveform?

Simply looking at the graph of the signal is rarely sufficient. Complicated waveforms often conceal their individual oscillatory components, making direct visual inspection unreliable.

To identify the hidden frequencies, we require a mathematical procedure capable of comparing an unknown signal with a known oscillation.

The development of such a comparison forms the central objective of this chapter.

Although this problem appears to belong exclusively to signal analysis, the underlying mathematics is surprisingly universal. The same geometric ideas used to compare ordinary vectors also govern the comparison of oscillatory functions, quantum states, and ultimately the Fourier transform itself.

4.1 How Do Physicists Compare Two Objects?

Before comparing oscillations, it is useful to recall a much simpler problem from elementary geometry.

Suppose we are given two vectors,

$$\mathbf{a} = (a_1, a_2)$$

and

$$\mathbf{b} = (b_1, b_2).$$

How can we determine whether these vectors point in similar directions?

Geometry answers this question through the dot product,

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2.$$

The dot product measures the degree of similarity between two vectors.

Large positive values indicate that the vectors point in nearly the same direction.

Negative values indicate opposite directions.

A value of zero signifies something particularly important.

The vectors are perpendicular, or orthogonal, to one another.

4.2 Orthogonality

The dot product possesses an especially important property.

It may be written in the equivalent form

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta,$$

where θ denotes the angle between the vectors.

If

$$\theta = 90^\circ,$$

then

$$\cos \theta = 0,$$

and consequently

$$\mathbf{a} \cdot \mathbf{b} = 0.$$

Two vectors satisfying this condition are said to be orthogonal.

Orthogonality implies that the vectors possess no component along one another. In other words, knowledge of one vector provides no information about the other.

This simple geometric idea lies at the heart of Fourier analysis.

4.3 Can Oscillations Behave Like Vectors?

The geometric ideas developed above naturally suggest a remarkable possibility.

Suppose that oscillations behave like vectors.

If this were true, one could compare an unknown signal with a known oscillation in exactly the same manner that one compares two ordinary vectors.

However, an immediate difficulty appears.

Vectors consist of only a finite collection of numbers.

Oscillations are continuous functions of time.

The ordinary dot product can therefore no longer be used directly.

A new mathematical operation is required—one capable of comparing entire functions rather than finite-dimensional vectors.

This operation is known as the inner product.

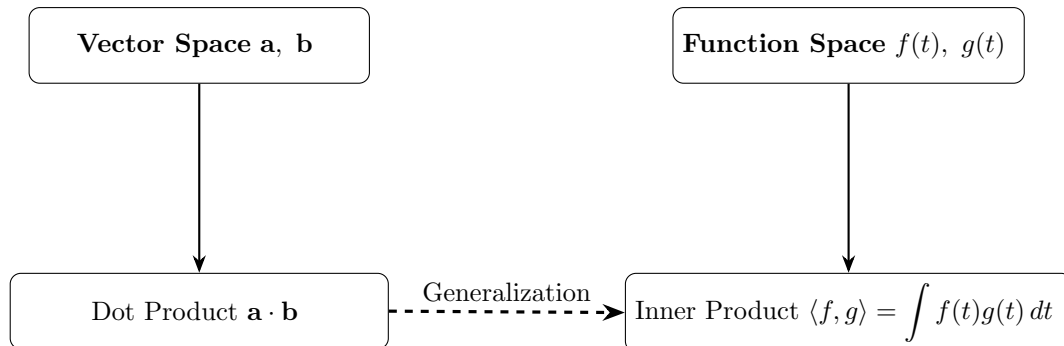


Figure 4.1: The inner product is the natural generalization of the familiar dot product. Just as the dot product compares vectors in ordinary Euclidean space, the inner product compares functions in an infinite-dimensional function space. This idea forms the mathematical bridge between elementary geometry and Fourier analysis.

4.4 From the Dot Product to the Inner Product

Consider two functions,

$$f(t)$$

and

$$g(t).$$

Unlike ordinary vectors, these objects possess infinitely many values because they are defined continuously over time.

To compare them, we generalize the ordinary dot product by replacing the finite summation with an integral over the entire interval of interest.

The resulting quantity,

$$\langle f, g \rangle = \int f(t)g(t) dt,$$

is known as the inner product.

Just as the dot product measures the similarity between vectors, the inner product measures the similarity between functions.

If the value of the inner product is large, the functions resemble one another.

If the value is zero, the functions are orthogonal.

4.5 Orthogonality of Oscillatory Functions

Having generalized the dot product to continuous functions, we are now in a position to compare different oscillatory modes.

Consider two cosine functions having angular frequencies ω_1 and ω_2 ,

$$f(t) = \cos(\omega_1 t),$$

and

$$g(t) = \cos(\omega_2 t).$$

Their similarity is measured through the inner product

$$\langle f, g \rangle = \int_0^T \cos(\omega_1 t) \cos(\omega_2 t) dt.$$

If the two oscillations possess different frequencies, the positive and negative contributions cancel one another over a complete interval, giving

$$\langle f, g \rangle = 0.$$

The two oscillations are therefore said to be *orthogonal*.

This result carries profound physical significance.

Two oscillations of different frequencies are mathematically independent. One frequency carries no information about another, just as two perpendicular vectors possess no component along one another.

Orthogonality is precisely the property that allows us to separate a complicated signal into its individual frequency components.

4.6 Projecting an Unknown Signal

Suppose an unknown acoustic signal

$$x(t)$$

has been recorded by the microphone.

Our objective is no longer to study a pure oscillation. Instead, we wish to determine whether a particular frequency is present within the unknown waveform.

To accomplish this, we compare the signal with a single oscillatory basis function,

$$\cos(\omega t).$$

The comparison is performed using the inner product,

$$\langle x(t), \cos(\omega t) \rangle.$$

If the value of this inner product is large, the signal possesses a strong component oscillating at frequency ω .

If the value is zero, that oscillation contributes nothing to the signal.

Thus, the inner product measures the contribution of each oscillatory mode to the construction of the unknown waveform.

The problem of signal analysis has therefore been transformed into a problem of projection.

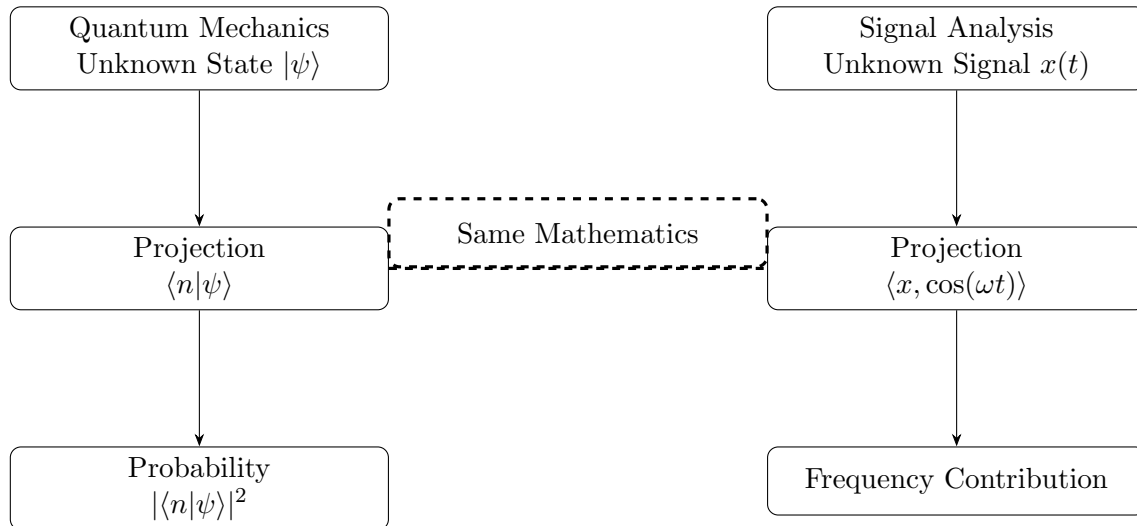


Figure 4.2: The mathematical structure of Fourier analysis mirrors the projection of quantum states onto an orthonormal basis. In both cases, projection coefficients reveal the contribution of elementary basis functions to an unknown object.

4.7 A Remarkable Analogy with Quantum Mechanics

The mathematical ideas developed above are not unique to signal analysis.

An almost identical construction appears in quantum mechanics.

A quantum state

$$|\psi\rangle$$

may be expanded in terms of a complete set of orthonormal basis states,

$$|n\rangle.$$

The quantity

$$\langle n|\psi\rangle$$

represents the projection of the unknown state onto the basis vector $|n\rangle$.

Its squared magnitude,

$$|\langle n|\psi\rangle|^2,$$

gives the probability of observing the system in that particular state.

The philosophy underlying Fourier analysis is strikingly similar.

Instead of projecting an unknown quantum state onto quantum basis vectors, we project an unknown signal onto oscillatory basis functions.

The resulting coefficients quantify the contribution of each frequency to the measured waveform.

The mathematics is identical.

Only the physical interpretation changes.

Looking Ahead

The central idea of this chapter may now be summarized in a single sentence.

To understand an unknown signal, we compare it with a complete collection of orthogonal oscillatory basis functions.

The remaining question is purely mathematical.

How can these projections be calculated for every possible frequency simultaneously?

The answer is provided by the Fourier transform.

In the next chapter, we shall derive this remarkable mathematical operation directly from the ideas of orthogonality and projection developed throughout the present chapter.

Chapter 5

The Fourier Transform as a Projection

The previous chapter established that oscillatory functions form an orthogonal basis and that the inner product provides a natural method for comparing an unknown signal with an individual oscillatory mode.

A fundamental question nevertheless remains unanswered.

Suppose that an unknown acoustic signal has been received,

$$x(t).$$

How can we determine not merely the contribution of one particular frequency, but the contribution of every physically possible frequency?

The answer is remarkably simple.

Instead of projecting the signal onto a single oscillatory basis function, we project it onto every oscillation simultaneously.

The collection of all such projection coefficients forms what is known as the *Fourier transform*.

5.1 From One Projection to Many

Suppose we compare the unknown signal with a single oscillation,

$$\cos(\omega t).$$

The corresponding inner product tells us how strongly that particular frequency contributes to the signal.

However, Nature imposes no restriction that the signal should contain only one oscillatory mode.

It may contain two frequencies.

Ten frequencies.

Or even an infinite continuum of frequencies.

Consequently, comparing the signal with only a single oscillation is insufficient.

The comparison must instead be repeated for every possible value of the angular frequency ω .

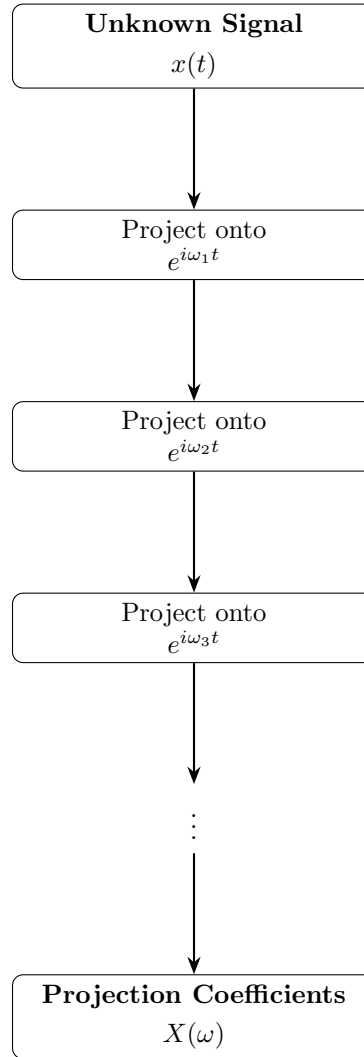


Figure 5.1: Conceptual illustration of the Fourier transform. The unknown signal is projected successively onto oscillatory basis functions of different angular frequencies. The collection of all projection coefficients forms the Fourier transform, $X(\omega)$.

5.2 Projection onto a Single Oscillation

The contribution of a single oscillatory mode is obtained by evaluating the inner product between the signal and the corresponding basis function,

$$\langle x(t), e^{i\omega t} \rangle.$$

Using the definition of the inner product developed in the previous chapter, this projection becomes

$$\int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt.$$

Notice that the exponential basis function now replaces the cosine representation introduced earlier.

Euler's relation,

$$e^{i\omega t} = \cos(\omega t) + i \sin(\omega t),$$

shows that the complex exponential simultaneously contains both sine and cosine oscillations.

It therefore forms a more complete basis for describing arbitrary wave phenomena.

What should we call this projection?

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt \quad (5.1)$$

We define this quantity as the Fourier Transform of the signal.

$$\boxed{X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt}$$

The function

$$X(\omega)$$

is not another signal.

It is a map of the contribution of every angular frequency contained within the original waveform.

Where $X(\omega)$ is large, the corresponding frequency is strongly represented.

Where it vanishes, that oscillatory mode is absent.

The Fourier transform therefore converts the question

“What happens at each instant of time?”

into the question

“Which frequencies are present?”

The Fourier transform therefore does not create new information.

It merely reveals information that was already present, but hidden within the time-domain waveform.

Time and frequency are not competing descriptions of a signal.

They are complementary mathematical representations of the same physical phenomenon.

Chapter 6

From Continuous Waves to Digital Measurements

Throughout the previous chapters, we have assumed that the acoustic signal is represented by a continuous function,

$$x(t),$$

defined for every instant of time.

This description is entirely appropriate from the standpoint of physics. Air pressure varies continuously, and the microphone converts these pressure fluctuations into a continuously varying electrical voltage.

However, a computer does not observe the world continuously.

Unlike a physicist equipped with mathematical analysis, a computer cannot evaluate a function at every instant of time. It possesses access only to a finite collection of numerical values stored in memory.

This immediately raises an important question.

How does a continuous physical wave become a sequence of numbers that a computer can analyze?

The answer lies in the process known as *sampling*.

6.1 The Process of Sampling

Suppose Rocky speaks into the microphone.

The acoustic pressure changes continuously with time, producing the function

$$x(t).$$

Instead of recording every instant, the computer measures the signal only at equally spaced moments,

$$t = 0, \Delta t, 2\Delta t, 3\Delta t, \dots$$

The result is no longer a continuous function.

Instead, the signal becomes the discrete sequence

$$x[0], x[1], x[2], x[3], \dots$$

where

$$x[n] = x(n\Delta t).$$

Thus, sampling converts a continuous physical measurement into a finite numerical representation suitable for digital computation.

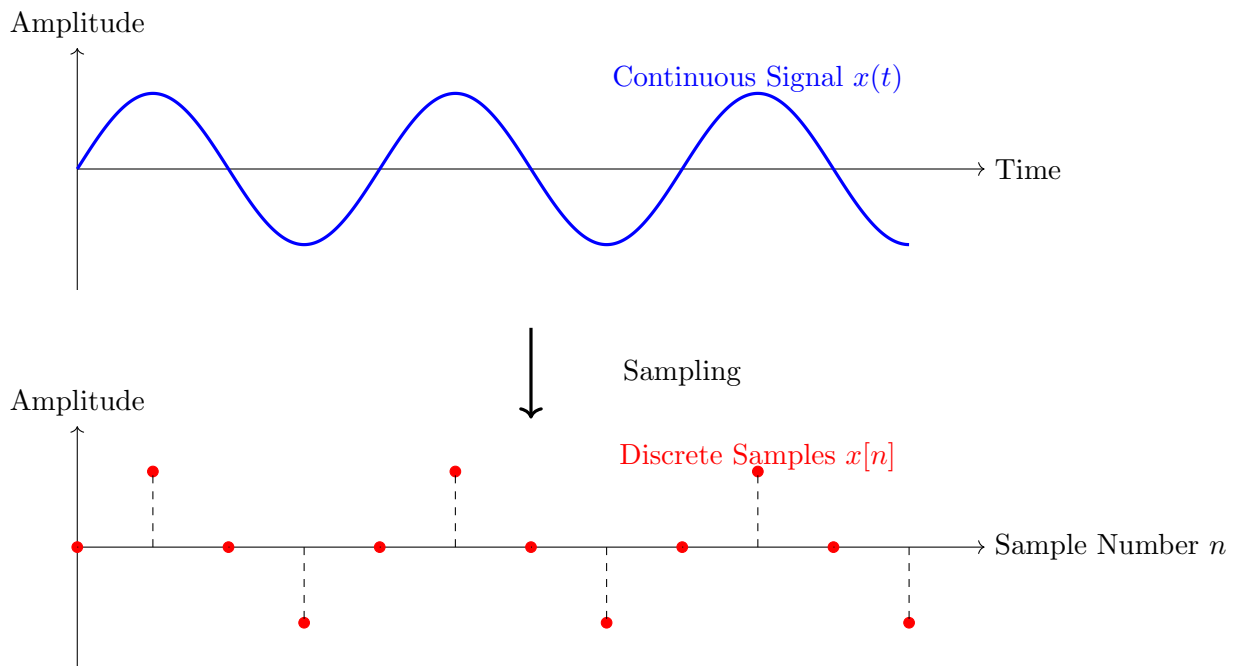


Figure 6.1: A microphone produces a continuous acoustic signal $x(t)$. During sampling, the computer measures the signal only at equally spaced instants, producing the discrete sequence $x[n]$. Although only a finite number of measurements are stored, these samples retain the information required for subsequent digital analysis.

6.2 Choosing the Sampling Rate

Sampling too slowly causes important details of the waveform to be lost.

Sampling unnecessarily fast increases storage requirements without providing significant additional information.

A natural question therefore arises.

How frequently should the signal be measured?

To answer this, let us consider the simplest possible oscillation,

$$x(t) = \cos(2\pi ft).$$

If only a few measurements are made during one oscillation, the computer cannot reliably

determine its frequency.

Consequently, the sampling rate must always exceed the oscillation frequency by a sufficient margin.

6.3 Why 44.1 kHz?

The process of sampling immediately raises an important practical question.

How frequently should the computer measure the incoming acoustic signal?

The answer depends entirely on the highest frequency that the signal may contain.

For human hearing, the audible frequency range extends approximately from

$$20 \text{ Hz} \leq f \leq 20\,000 \text{ Hz}.$$

If the sampling rate is chosen too low, the rapidly oscillating components of the waveform cannot be faithfully represented. On the other hand, sampling at an unnecessarily high rate increases storage requirements without providing significant additional physical information.

A natural compromise is therefore required.

According to the sampling criterion that will be discussed in the next section, the sampling frequency must exceed twice the highest frequency present in the signal.

Since the upper limit of human hearing is approximately

$$20 \text{ kHz},$$

the sampling frequency should therefore exceed

$$40 \text{ kHz}.$$

For this reason, compact discs and many digital audio systems adopt the standard sampling frequency

$$f_s = 44.1 \text{ kHz}.$$

This corresponds to

$$44\,100$$

measurements every second.

Equivalently, the time interval between two consecutive measurements is

$$\Delta t = \frac{1}{44100} \approx 22.7 \mu\text{s}.$$

Thus, every $22.7 \mu\text{s}$, the computer records one new value of the acoustic signal.

The continuous pressure wave produced by Rocky's voice has therefore been transformed into a sequence of numerical measurements suitable for digital analysis.

6.4 The Nyquist Criterion

Why must the sampling frequency exceed twice the highest frequency present in the signal?

To understand this requirement, consider a simple oscillation,

$$x(t) = \cos(2\pi ft).$$

If only one measurement is taken during each oscillation, the computer cannot determine whether the signal is increasing or decreasing between successive samples. In fact, many different oscillations may produce exactly the same sequence of sampled values.

The original frequency therefore becomes ambiguous.

To reconstruct an oscillation uniquely, the computer must record more than one sample during every cycle.

This requirement leads to the celebrated Nyquist criterion,

$$f_s > 2f_{\max},$$

where f_s denotes the sampling frequency and $f_{\max}$ represents the highest frequency contained within the signal.

Whenever this condition is satisfied, every oscillatory component of the waveform can, in principle, be reconstructed from the sampled data.

If the condition is violated, different frequencies become indistinguishable after sampling. This phenomenon is known as *aliasing*, in which high-frequency oscillations appear as lower-frequency components within the reconstructed signal.

The Nyquist criterion therefore provides the fundamental physical limit governing the conversion of continuous signals into digital measurements.

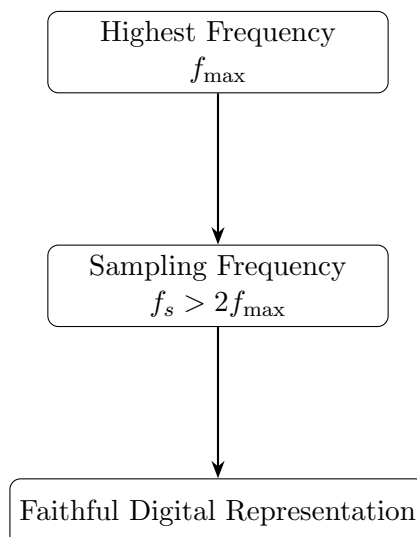


Figure 6.2: The Nyquist criterion ensures that the sampling frequency is sufficiently large to preserve the information contained in the original continuous signal.

6.5 From the Continuous Fourier Transform to the Discrete Fourier Transform

The Fourier transform developed in the previous chapter assumes that the signal is known continuously for every value of time.

Mathematically,

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-i\omega t} dt.$$

A digital computer, however, never possesses access to the continuous function $x(t)$.

Instead, it stores only the sampled sequence

$$x[0], x[1], x[2], \dots, x[N-1].$$

Consequently, the continuous integral appearing in the Fourier transform cannot be evaluated directly.

The only information available to the computer consists of a finite collection of sampled values.

The natural mathematical replacement for a continuous integral is therefore a finite summation.

Replacing the continuous variable t by the discrete index n , and replacing the integral by a summation, leads to

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-i2\pi kn/N}.$$

This equation is known as the *Discrete Fourier Transform* (DFT).

Its physical interpretation remains identical to that of the continuous Fourier transform.

Rather than projecting a continuous signal onto every oscillatory basis function, the DFT projects a finite collection of sampled measurements onto a discrete set of oscillatory basis functions.

The result,

$$X[k],$$

describes the contribution of each discrete frequency contained within the measured signal.

Thus, the DFT does not introduce a new physical principle.

It is simply the digital realization of the projection idea developed throughout the previous chapters.

Looking Ahead

The Discrete Fourier Transform provides the mathematical language through which computers analyze digital signals.

However, evaluating this summation directly becomes increasingly expensive as the amount of data grows.

Is there a more efficient way to perform exactly the same calculation?

The answer is the Fast Fourier Transform, one of the most influential numerical algorithms ever developed.

Chapter 7

From Frequencies to Meaning

The previous chapters have shown how a microphone records an acoustic signal, how that signal is represented mathematically, and how the Fourier transform reveals the frequencies hidden within it.

Suppose now that Grace has successfully analyzed Rocky's acoustic signal.

The waveform has been digitized.

The Fourier transform has been computed.

The dominant frequencies are known.

Yet one fundamental question remains unanswered.

How do frequencies become language?

Neither the Fourier transform nor the Fast Fourier Transform performs translation.

They merely reveal the physical structure of the received signal.

The problem of assigning meaning to that structure belongs to a different stage of analysis.

This chapter explores, at a conceptual level, how repeated frequency patterns may gradually evolve into words, vocabulary, and eventually an entire language.

The emphasis remains on the physical ideas rather than the computational details.

7.1 Why One Fourier Transform Is Not Enough

A single Fourier transform describes the frequency content of the entire recorded signal.

Suppose Rocky speaks continuously for ten seconds.

Computing one Fourier transform over the complete recording would reveal which frequencies are present during the entire interval.

However, language is dynamic.

Different sounds appear at different moments.

Some frequencies exist only briefly.

Others persist throughout the conversation.

Consequently, a single Fourier transform loses all information about when a particular frequency occurred.

To recover this temporal information, the recording must be divided into many short intervals, each of which is analyzed independently.

This idea forms the basis of the Short-Time Fourier Transform (STFT).

7.2 Listening Frame by Frame

Instead of analyzing the entire recording simultaneously, the signal is divided into many small segments called *frames*.

A typical speech-processing system uses frames of approximately

20–30 ms.

Suppose Rocky speaks continuously for ten seconds.

Rather than computing one Fourier transform,

$$x(t) \longrightarrow X(\omega),$$

Grace’s computer performs hundreds of smaller Fourier transforms,

$$\text{Frame}_1 \rightarrow X_1(\omega),$$

$$\text{Frame}_2 \rightarrow X_2(\omega),$$

$$\text{Frame}_3 \rightarrow X_3(\omega),$$

and so forth until the entire recording has been analyzed.

Each frame therefore produces its own frequency spectrum.

The computer now knows not only which frequencies exist, but also when they occur.

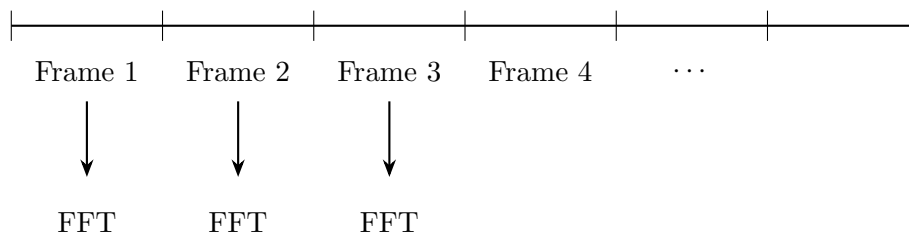


Figure 7.1: Instead of analyzing the entire recording simultaneously, the signal is divided into short frames. Each frame is analyzed independently using the Fourier transform, allowing the frequency content to be tracked as it evolves with time.

7.3 The Spectrogram

Once the Fourier transform has been computed for every frame, the results may be displayed graphically.

The resulting image is called a *spectrogram*.

Unlike the ordinary waveform, which shows amplitude as a function of time, the spectrogram simultaneously displays

- time,
- frequency,
- signal intensity.

Time is represented along the horizontal axis.

Frequency is represented along the vertical axis.

The brightness or colour at each point indicates the strength of that frequency during the corresponding time interval.

The spectrogram therefore provides a visual history of how the frequency content evolves throughout the recording.

For speech signals, repeated sounds often appear as repeated spectral patterns, making the spectrogram one of the most informative representations used in modern speech analysis.

7.4 A Toy Example Inspired by *Project Hail Mary*

Suppose that Grace has already computed the spectrogram of Rocky's speech.

For simplicity, imagine that only four dominant frequencies are observed,

$$400 \text{ Hz}, \quad 600 \text{ Hz}, \quad 800 \text{ Hz}, \quad 900 \text{ Hz}.$$

Further suppose that repeated observations reveal the following patterns.

Dominant Frequencies	Observation
400 Hz + 600 Hz	Rocky points to himself
800 Hz + 900 Hz	Rocky requests water
400 Hz + 600 Hz + 800 Hz + 900 Hz	Rocky says the complete phrase

Initially, Grace has no knowledge of Rocky's language.

She merely records repeated associations between acoustic patterns and observable events.

After many observations, the computer begins forming statistical hypotheses.

The repeated occurrence of

$$(400, 600)$$

whenever Rocky refers to himself suggests that this frequency combination may correspond to the concept

“I”.

Similarly,

$$(800, 900)$$

becomes associated with the concept

“water”.

Eventually, when both patterns appear consecutively,

$$(400, 600) \rightarrow (800, 900),$$

the computer proposes the interpretation

I need water.

Of course, this example is intentionally simplified.

Real language learning requires many thousands of observations, together with sophisticated statistical and machine-learning algorithms.

Nevertheless, the underlying philosophy remains identical.

The computer never begins with a dictionary.

It begins with repeated physical observations and gradually discovers structure through statistical regularities.

Moving ahead there is a vast field of Machine Learning and AI which recognises the pattern and behaviours to build grammar, vocabulary and a complete Dictionary to transcribe Rocky’s Language to English by Grace.

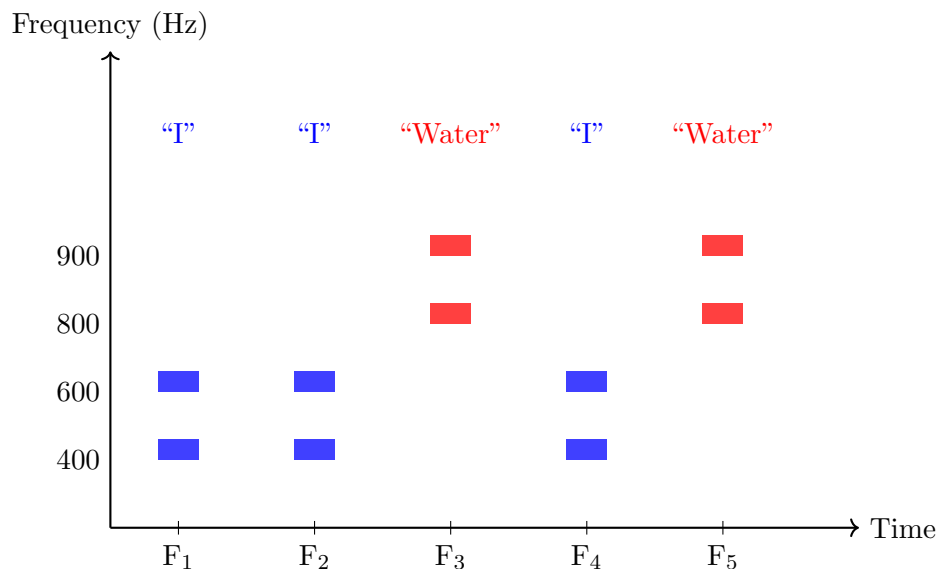


Figure 7.2: A schematic spectrogram illustrating the evolution of frequency content with time. Each vertical slice corresponds to the Fourier transform of one short frame. Repeated frequency combinations produce repeated spectral patterns. In this simplified example, the frequency pair (400 Hz, 600 Hz) is repeatedly associated with one acoustic unit, while (800 Hz, 900 Hz) forms another. Such recurring patterns provide the statistical structure upon which higher-level interpretation may eventually be built.

Figure 7.2 illustrates the essential idea behind a spectrogram. Rather than displaying the acoustic waveform directly, the spectrogram represents how the frequency content evolves with time.

Each column corresponds to one short frame of the recording. Within that frame, the Fourier transform identifies the dominant frequency components. The brightness (represented schematically here by coloured blocks) indicates the presence of those frequencies during the corresponding time interval.

Notice that the frequency pair

(400 Hz, 600 Hz)

appears repeatedly in several frames, whereas the pair

(800 Hz, 900 Hz)

appears in others.

A physicist would simply conclude that these are recurring acoustic structures. Whether these structures correspond to words, syllables, or other linguistic units cannot be determined by Fourier analysis alone. Nevertheless, identifying such repeated patterns is the indispensable first step toward any subsequent interpretation.